Epileptic Seizure State Classification Using Quantum Inspired Techniques

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Abstract

Accurate discrimination between preictal and ictal states from electroencephalography (EEG) signals is essential for epilepsy management and represents a key step toward reliable seizure monitoring and early warning systems. However, the complex, non-stationary, and highly patient-specific nature of EEG signals continues to limit the discriminative capability of conventional feature representations. This study investigates whether quantum-inspired (QI) descriptors can provide a more informative and robust feature space for epileptic seizure state classification. Seizure state discrimination is evaluated across three feature domains: time-domain features, frequency-domain features, and quantum-inspired descriptors designed to capture global signal structure and energy redistribution characteristics. Classification is performed using a support vector machine (SVM) with an RBF kernel under a strict patient-independent evaluation protocol on the CHB-MIT Scalp EEG dataset, enabling assessment of generalization to unseen subjects. Experimental results demonstrate that quantum-inspired features achieve stronger discriminative performance than individual time-domain or frequency-domain representations, and provide performance comparable to their classical combination. Furthermore, integrating time-domain, frequency-domain, and quantum-inspired features yields the best overall performance, achieving an ROC–AUC of 0.9376 on unseen patients, with overall accuracy remaining consistently high across configurations. These findings suggest that quantum-inspired feature representations capture complementary system-level information that enhances seizure state classification when combined with conventional descriptors.

Keywords: Epileptic seizure classification, Electroencephalography, Quantum-inspired features, Patient-independent evaluation, Support vector machine

1 Introduction

Epilepsy is a chronic neurological disorder characterized by recurrent seizures arising from abnormal and excessive neuronal synchronization in the brain. Affecting more than 50 million individuals worldwide, it represents a major public health challenge with significant clinical and socio-economic impact [1]. The reliable analysis of seizure-related brain states is therefore a critical requirement for effective epilepsy management and the development of automated seizure monitoring systems.

Electroencephalography (EEG) remains the principal non-invasive technique for recording epileptic brain activity due to its high temporal resolution and widespread clinical use [2]. Nevertheless, EEG signals are inherently non-stationary, noisy, and highly variable across patients. Furthermore, epileptic activity evolves through multiple physiological states, including the preictal phase preceding seizure onset and the ictal phase itself. Accurate discrimination between these states is particularly challenging under patient-independent evaluation, where inter-subject variability further limits model generalization.

Most existing seizure analysis approaches rely on handcrafted time-domain and frequency-domain features that effectively describe localized signal characteristics [3]. However, such representations may be limited in capturing global signal structure, nonlinear dependencies, and collective dynamics across EEG channels. To address this, this study investigates the use of *quantum-inspired (QI)* feature representations derived from principles of quantum information theory. By modeling EEG segments as interdependent systems and capturing entropy-based complexity and global correlations, QI features are evaluated alongside conventional descriptors within a strict patient-independent framework, with the objective of improving epileptic seizure state classification.

2 Related Work

Epileptic seizure analysis has evolved from traditional handcrafted statistical approaches to data-driven machine learning and deep learning methods, with several studies integrating both paradigms [3, 4]. Conventional time-domain and frequency-domain features, such as statistical moments and spectral band power measures [5], remain widely used due to their interpretability and computational efficiency. However, these representations primarily capture localized signal characteristics and may be limited in modeling global structure and nonlinear dependencies inherent in multichannel EEG signals.

Recent advances in deep learning have demonstrated improved performance in EEG-based seizure analysis. Architectures combining convolutional neural networks (CNNs) and recurrent models such as bidirectional long short-term memory (BiLSTM) networks have been shown to enhance spatial and temporal feature representation [6–8]. Attention mechanisms further improve model focus and interpretability by emphasizing seizure-relevant temporal patterns [9]. While these approaches report strong performance under various experimental settings, they predominantly address seizure versus non-seizure classification and often rely on patient-dependent or partially overlapping evaluation protocols.

Discriminating between preictal and ictal states under strictly patient-independent evaluation remains comparatively underexplored and poses additional challenges due to temporal proximity and gradual state transitions[10]. To address this gap, recent research has begun exploring feature-level enhancements beyond conventional descriptors. Quantum-inspired (QI) techniques, motivated by concepts from quantum information theory, aim to capture global correlations, synchrony, and entropy-based complexity in EEG signals [11–13]. However, the effectiveness of such QI representations for preictal–ictal discrimination, particularly within a patient-independent classification framework, has not been thoroughly investigated. This study addresses this limitation by systematically evaluating quantum-inspired features alongside classical time- and frequency-domain representations.

3 Methodology

This section describes the dataset, preprocessing pipeline, feature engineering strategy, experimental setup, and classification framework adopted for epileptic seizure state classification. All parameters and evaluation protocols are explicitly specified to ensure reproducibility and to support strict patient-independent assessment.

3.1 Dataset

Experiments were conducted using the CHB–MIT Scalp EEG database [14, 15], which consists of long-term scalp EEG recordings from pediatric subjects with intractable epilepsy. The recordings were acquired at a sampling rate of 256 Hz using a standard 10–20 electrode configuration, with clinically annotated seizure onset and offset times provided in accompanying summary files.

To evaluate generalization across subjects, a strict patient-independent protocol was adopted. EEG data from patients `chb01` to `chb20` were used for training, while patients `chb21`, `chb22`, and `chb23` were reserved exclusively for testing. No data from test subjects were used during feature normalization, model fitting, or parameter selection.

3.2 Preprocessing

EEG preprocessing was performed to reduce noise, artifacts, and inter-recording variability. All channels were band-pass filtered between 1 and 40 Hz and notch filtered at 50 Hz to suppress power-line interference. Independent Component Analysis (ICA) was applied to attenuate ocular and muscular artifacts when channel dimensionality permitted [16].

To ensure a consistent channel layout across recordings with variable montages, a fixed set of 23 EEG channels was selected based on their prevalence across the dataset. Channels absent in individual recordings were zero-padded to preserve a uniform channel ordering.

The preprocessed EEG signals were segmented into fixed-length windows of 1 second duration (256 samples) using a stride of 2 seconds to reduce temporal redundancy and limit correlation between adjacent samples during evaluation. For each clinically

annotated seizure, windows were labeled as *preictal* if they occurred between 60 s and 300 s prior to seizure onset, and as *ictal* if they overlapped the seizure interval. A minimum inter-seizure separation of 10 minutes was enforced, and seizures violating this constraint were excluded to prevent temporal label contamination.

3.3 Feature Engineering

Each EEG window was represented using handcrafted features extracted independently from three complementary domains: time-domain, frequency-domain, and quantum-inspired representations. Feature extraction was performed at the window level without temporal smoothing or post hoc aggregation to evaluate intrinsic feature-level discriminability between seizure states.

Time-Domain Features

Time-domain features were extracted to characterize amplitude dynamics, signal variability, and cross-channel statistical structure of EEG windows. Features were drawn from standard statistical feature families commonly used in EEG analysis, including global signal statistics computed by pooling samples across channels, as well as channel-wise descriptors summarized across channels. These included measures of central tendency, dispersion, energy, peak-to-peak amplitude, zero-crossing behavior, and Hjorth activity, mobility, and complexity[17]. In addition, inter-channel linear dependence was quantified using summary statistics of pairwise Pearson correlations. Collectively, these feature families yielded 29 time-domain features per EEG window. These descriptors rely on explicit statistical aggregation and do not provide a unified system-level representation of multichannel EEG activity.

Frequency-Domain Features

Frequency-domain features were computed using Welch's method for power spectral density estimation[5]. Features were derived from standard spectral feature families, including absolute and relative band power measures, band power ratios, and entropy-based spectral descriptors computed over canonical EEG frequency bands. These features summarize rhythmic activity and frequency-specific energy redistribution associated with preictal–ictal transitions. In total, 17 frequency-domain features were extracted per EEG window.

Quantum-Inspired Features

Quantum-inspired (QI) features were designed to capture global inter-channel structure and system-level dynamics that are not explicitly modeled by conventional time- or frequency-domain descriptors. In this study, the term *quantum-inspired* refers to feature representations motivated by concepts from quantum information theory and quantum phase-space analysis, while being computed entirely on classical EEG signals [12, 18].

For each EEG window, multichannel activity was represented using a density matrix ρ, defined in the standard quantum-mechanical form as [18]

$$\rho = \sum_i p_i \, |\psi_i\rangle\langle\psi_i|, \quad \sum_i p_i = 1, \tag{1}$$

where $|\psi_i\rangle$ denotes a normalized multichannel EEG state vector formed from windowed channel amplitudes, and p_i represents an energy-normalized weighting derived from channel-wise signal power. In this representation, diagonal elements of ρ encode normalized channel energy, while off-diagonal elements quantify inter-channel coupling, yielding a compact system-level description of EEG dynamics [13].

From the resulting density matrix, information-theoretic descriptors were derived to quantify global disorder, synchronization, and complexity of multichannel EEG activity, including von Neumann entropy, purity, linear entropy, quantum coherence, quantum Fisher information, and half-system entanglement entropy computed by partitioning channels into equal subsets [19, 20].

In addition, phase-space characteristics were captured using the Wigner–Ville distribution, a time–frequency representation with origins in quantum mechanics. Since the Wigner–Ville distribution is a quasi-probability representation that may assume negative values due to cross-term interference, entropy- and moment-based descriptors were computed using an energy-normalized magnitude representation to ensure non-negativity and numerical stability, thereby enabling a valid probabilistic interpretation for entropy computation [21, 22]. From the resulting distribution, five statistical descriptors—energy, entropy, variance, skewness, and kurtosis—were extracted per window to characterize non-stationary oscillatory structure and spectral redistribution during seizure transitions.

In total, 11 quantum-inspired features were extracted per EEG window, comprising 6 density-matrix-based descriptors and 5 Wigner-based phase-space descriptors. The complete set of quantum-inspired features used in this study is summarized in Table 1.

3.4 Experimental Setup

To assess the discriminative capability of different feature representations, five experimental configurations were evaluated: (i) time-domain features only, (ii) frequency-domain features only, (iii) combined time–frequency features, (iv) quantum-inspired features only, and (v) a combined feature set integrating time-domain, frequency-domain, and quantum-inspired representations.

All models were trained using EEG windows from patients `chb01`–`chb20` and evaluated exclusively on unseen patients `chb21`, `chb22`, and `chb23`, ensuring a strict patient-independent assessment. Feature normalization parameters were learned solely from the training data and applied unchanged to the test set. No temporal smoothing or post-processing was applied during training or evaluation, allowing performance differences to reflect the intrinsic discriminative capability of the extracted features.

Table 1 Quantum-inspired EEG features used in this study

Feature	Mathematical Definition	Interpretation in EEG Context
Von Neumann Entropy	$S(\rho) = -\mathrm{Tr}(\rho \log \rho)$ [13, 18]	Global disorder and complexity of multichannel EEG dynamics
Purity	$\mathcal{P}(\rho) = \mathrm{Tr}(\rho^2)$ [18]	Degree of global synchronization and source dominance
Linear Entropy	$S_L(\rho) = 1 - \mathrm{Tr}(\rho^2)$ [11]	Approximate mixedness of multichannel EEG activity
Quantum Coherence	$C(\rho) = \sum_{i \neq j} \lvert\rho_{ij}\rvert$ [19]	Strength of inter-channel coupling and coordinated activity
Quantum Fisher Information	$F_Q(\rho) = 2\sum_{i,j} \frac{(\lambda_i - \lambda_j)^2}{\lambda_i + \lambda_j} \lvert\langle i\rvert G \rvert j\rangle\rvert^2$ [20]	Sensitivity of global EEG structure to perturbations under a fixed generator
Entanglement Entropy (Half-System)	$S(\rho_A) = -\mathrm{Tr}(\rho_A \log \rho_A)$ [12, 18]	Information coupling between EEG channel subsets
Wigner Energy	$E_W = \sum_{t,f} \lvert W(t,f)\rvert^2$ [21]	Overall energy concentration in the time–frequency plane
Wigner Entropy	$S_W = -\sum_{t,f} \tilde{W}(t,f) \log \tilde{W}(t,f)$ [21]	Time–frequency complexity and spectral dispersion
Wigner Variance	$\mathrm{Var}(\tilde{W}) = \mathbb{E}[(\tilde{W} - \mu_{\tilde{W}})^2]$ [22]	Spread of energy in the phase-space representation
Wigner Skewness	$\mathrm{Skew}(\tilde{W}) = \mathbb{E}[(\tilde{W} - \mu_{\tilde{W}})^3]/\sigma^3_{\tilde{W}}$ [22]	Asymmetry of transient oscillatory patterns
Wigner Kurtosis	$\mathrm{Kurt}(\tilde{W}) = \mathbb{E}[(\tilde{W} - \mu_{\tilde{W}})^4]/\sigma^4_{\tilde{W}}$ [22]	Peakedness and intermittency of phase-space structure

3.5 Model and Training

Seizure state classification was performed using a support vector machine (SVM) with a radial basis function (RBF) kernel. All features were standardized using z-score normalization computed solely from the training data, and class weighting was incorporated into the SVM objective to address imbalance between preictal and ictal samples. To ensure a fair comparison across feature configurations, model hyperparameters were kept fixed throughout all experiments, with the regularization parameter set to $C = 1.0$ and the RBF kernel width determined using the `scale` heuristic. Probabilistic outputs were enabled to facilitate ROC–AUC computation, and model performance was evaluated exclusively on unseen patients using the area under the receiver operating characteristic curve as the primary metric.

4 Results

Five experimental configurations were evaluated to assess the discriminative capability of different feature representations: time-domain features, frequency-domain features, quantum-inspired features, a combined time–frequency feature set, and a unified representation integrating all feature domains, with a summary of class-wise performance reported in Table 2.

Table 2 Class-wise performance comparison across feature representations

Feature Set	Preictal		Ictal		ROC–AUC
	Precision	Recall	Precision	Recall	
Time-domain only	0.94	0.94	0.70	0.71	0.8890
Frequency-domain only	0.94	0.94	0.70	0.73	0.9078
Time + Frequency	0.94	0.88	0.64	0.81	0.9151
Quantum-inspired only	0.97	0.78	0.50	0.90	0.9239
Time + Frequency + Quantum	0.93	0.95	0.78	0.72	0.9376

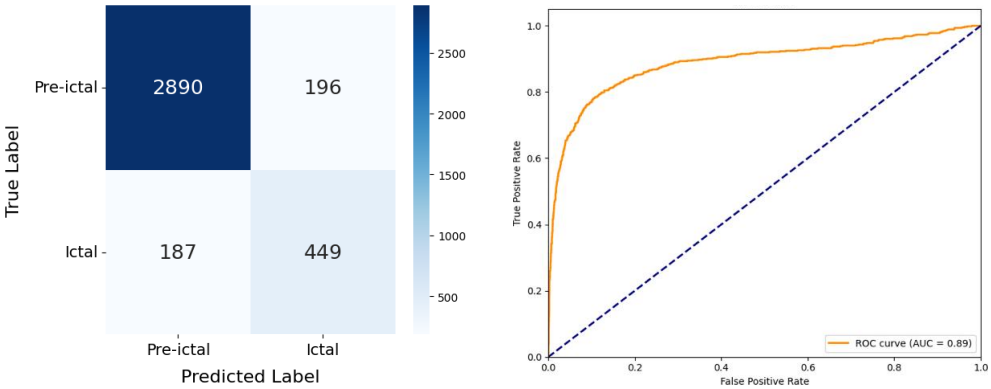

Fig. 1 Confusion matrix (left) and ROC curve (right) for the time-domain feature model.

Using time-domain features alone, the classifier achieved an ROC–AUC of 0.8890 on the test set. The preictal class showed strong performance, with both precision and recall of 0.94, while the ictal class achieved a precision of 0.70 and a recall of 0.71 (Figure 1). When frequency-domain features were used, performance improved modestly, yielding an ROC–AUC of 0.9078. Preictal precision and recall remained at 0.94, whereas ictal precision and recall increased slightly to 0.70 and 0.73, respectively (Figure 2).

Using quantum-inspired features alone (Table 1), the classifier achieved a higher ROC–AUC of 0.9239. In this setting, the preictal class attained a precision of 0.97 but a lower recall of 0.78, while the ictal class exhibited a precision of 0.50 and a notably higher recall of 0.90 (Figure 4). This indicates a shift in sensitivity toward ictal detection when using quantum-inspired representations.

Combining time-domain and frequency-domain features resulted in an ROC–AUC of 0.9151. The preictal class achieved a precision of 0.94 and a recall of 0.88, while ictal precision and recall were 0.64 and 0.81, respectively (Figure 3). The unified feature representation integrating time-domain, frequency-domain, and quantum-inspired features achieved the highest overall performance, with an ROC–AUC of 0.9376. In this configuration, preictal precision and recall were 0.93 and 0.95, respectively, while ictal precision and recall reached 0.78 and 0.72 (Figure 5).

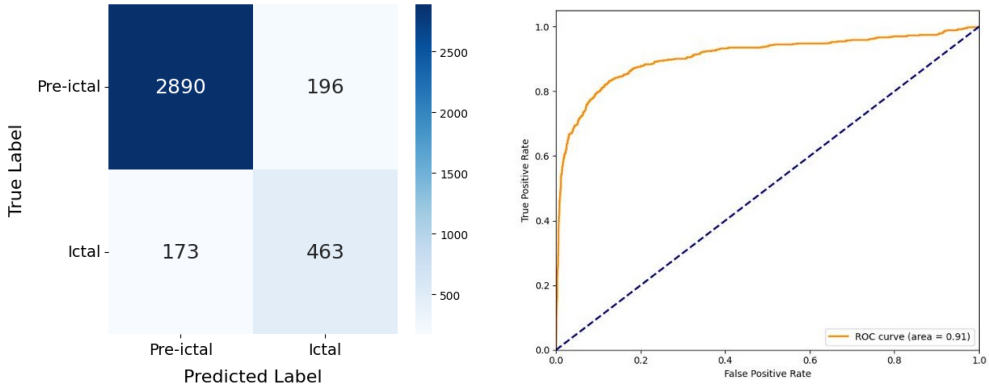

Fig. 2 Confusion matrix (left) and ROC curve (right) for the frequency-domain feature model.

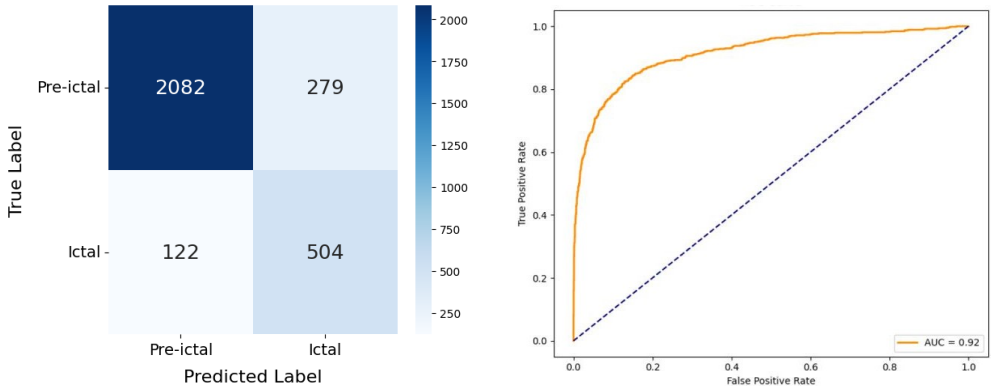

Fig. 3 Confusion matrix (left) and ROC curve (right) for the time–frequency feature model.

5 Discussion

This study systematically evaluated classical and quantum-inspired feature representations for preictal–ictal seizure state classification under a strict patient-independent protocol. The results demonstrate that feature representation plays a decisive role in determining generalization performance on unseen subjects, even when the same classifier and training procedure are used.

Conventional time-domain features achieved strong baseline performance, reflecting their ability to capture localized amplitude variations associated with seizure activity. However, their discriminative capability is limited to short-term statistical characteristics and does not explicitly encode global inter-channel relationships. Frequency-domain features provided a modest improvement, indicating that spectral energy redistribution across canonical EEG bands contains additional information relevant to seizure transitions.

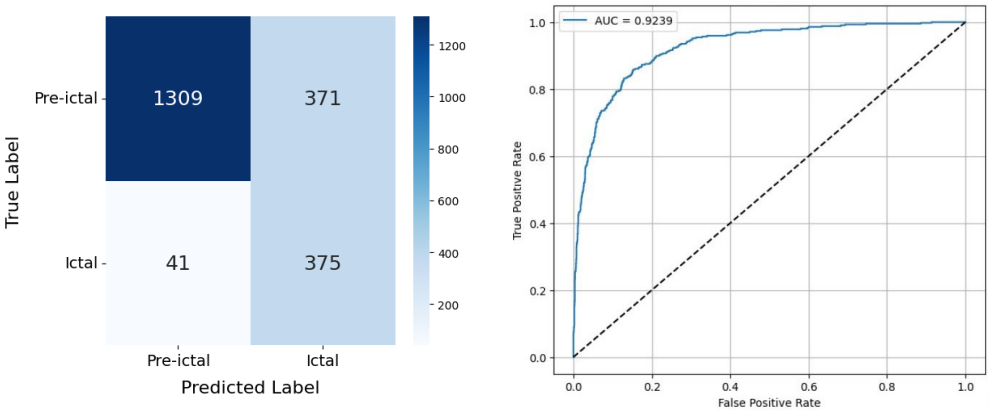

Fig. 4 Confusion matrix (left) and ROC curve (right) for the quantum-inspired feature model.

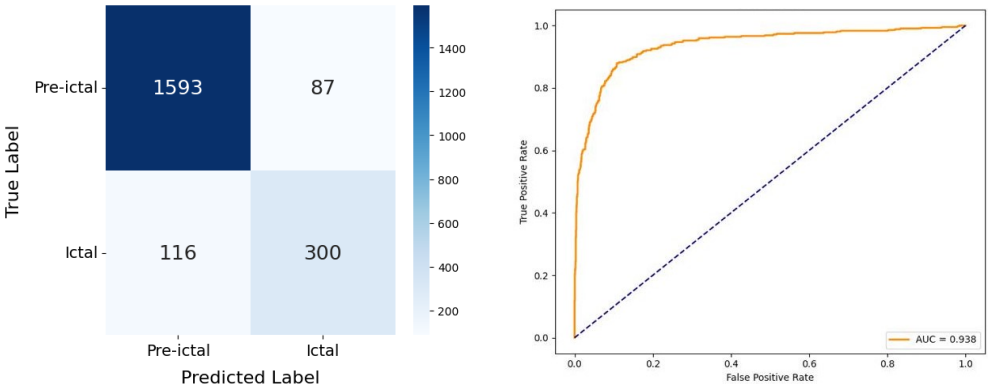

Fig. 5 Confusion matrix (left) and ROC curve (right) for the combined feature model.

Combining time-domain and frequency-domain features further improved performance, confirming that temporal and spectral descriptors offer complementary perspectives on seizure dynamics. The most notable improvement, however, was observed with quantum-inspired features. When evaluated independently, these descriptors achieved higher ROC–AUC than all classical baselines, highlighting their ability to capture system-level structure that is not explicitly represented by localized statistical or spectral measures. Notably, quantum-inspired features exhibited increased sensitivity to ictal events, reflected in higher recall, albeit with reduced precision when used in isolation.

The superior performance of quantum-inspired features can be attributed to their global formulation. By modeling multichannel EEG windows using density matrices, these features encode inter-channel coupling, synchronization, and energy redistribution in a unified representation. Entropy- and coherence-based descriptors quantify global complexity and coordination, while Wigner-based phase-space features capture transient oscillatory structure and non-stationary spectral behavior that are difficult to characterize using conventional power-based descriptors alone.

The best overall performance was achieved by integrating time-domain, frequency-domain, and quantum-inspired features. This result indicates that quantum-inspired descriptors do not replace classical features but instead complement them by capturing higher-order structure, while classical features continue to represent localized amplitude and spectral variations. Their integration yields a richer and more discriminative feature space, leading to improved robustness under patient-independent evaluation.

Importantly, all experiments were conducted without temporal smoothing, heuristic post-processing, or patient-specific tuning. Performance gains therefore arise from intrinsic feature discriminability rather than decision-level aggregation or temporal bias, strengthening the relevance of the findings for practical seizure monitoring systems. While quantum-inspired feature extraction introduces additional computational complexity, all features are computed on classical hardware and remain tractable at the window level.

6 Conclusion

This study investigated epileptic seizure state classification using classical and quantum-inspired feature representations under a strict patient-independent evaluation framework. The results show that quantum-inspired features outperform conventional time-domain and frequency-domain descriptors when used independently and, more importantly, that their integration with classical features yields the best overall performance, achieving an ROC–AUC of 0.9376 on unseen patients. These findings indicate that quantum-inspired representations capture complementary, system-level information beyond localized statistical and spectral characteristics, enhancing robustness and generalization across subjects. By adopting a transparent SVM-based classification framework and avoiding temporal smoothing or patient-specific optimization, the proposed approach emphasizes intrinsic feature discriminability and reproducibility. Overall, the hybrid feature representation offers a promising direction for clinically reliable seizure monitoring and early warning systems, with future work focused on real-time deployment and extension to multimodal neurophysiological data.

Declarations

The authors declare no competing interests. This study used publicly available, anonymized EEG data from the CHB–MIT Scalp EEG Database; no new data were collected and no experiments were performed on human subjects by the authors, and ethical approval was therefore not required. Informed consent was obtained by the original data collectors. The data used in this study are available from the PhysioNet repository, and the code used for feature extraction and model training is available from the corresponding author upon reasonable request. This research received no external funding.

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